Stage 1: Data Preparation

GitSkills
SQLite

Scope filter
SKILL.md, 200+ chars

Channels
name, prose, code

Repo context
owner, first commit

Extracted
1,586,429 skills (sample: 11,009)

skills

Stage 2: Candidate Pair Generation

MinHash LSH
J ≥ 0.3

Same name
≤ 45 per name

LSA (sample only)
cos ≥ 0.9

Union, bands
B1–B6

Candidates
9.0M pairs (full release)

candidate pairs

Stage 3: Manual Analysis

Explore
read all bands

Sample
400 pairs, 7 strata

Annotate
2 per pair

Agreement
Cohen's κ

Synthesize
final rubric

Gold set
labelled pairs

gold set

all candidate pairs

Stage 4: Model Construction and Evaluation

Features
19 in 6 groups

CV split
5-fold grouped

Training
LR / RF / GB

Evaluation
vs 4 baselines

Ablation
drop one group

RQ1, RQ2
best model

trained model

Stage 5: Lineage Recovery and Reuse Analysis

Phase 4: copy network
planned

RQ3 reuse
exact vs family

Hidden reuse
no shared hash or name

Louvain
families

Graph
derived edges

Score pairs
all candidates; 120 checked by hand